Table 36. Reference implementation modules

Module	What it does	Requirements	Tests
bapil.text	Unicode normalisation for Bangla, digit folding, light Bangla suffix stripping, bilingual glossary expansion	FR-01, NFR-01	test_text
bapil.corpus	Loads Markdown documents, splits them into cited passages, records provenance and SHA-256 checksums	FR-03	test_corpus
bapil.search	BM25 index over passages with language-aware tokenisation and cross-lingual query expansion	FR-01	test_search
bapil.answer	Extractive answers in which every sentence carries a citation; abstains below an evidence threshold	FR-02, NFR-08	test_answer
bapil.risk	Risk-tier questionnaire and scoring rules of Annex B, returning tier, reasons and required controls	FR-07, GR-02, GR-06	test_risk
bapil.registry	AI system register validated against a JSON schema; exports the public transparency register	FR-06, FR-09	test_registry
bapil.evaluate	Recall@k, MRR and nDCG over a versioned question set; JSON report	FR-08	test_evaluate
bapil.audit	Hash-chained, tamper-evident JSONL audit log with verification	GR-04	test_audit
bapil.api	Minimal JSON HTTP API (search, answer, risk) on the Python standard library	FR-10	test_api
bapil.cli	Command line: index, search, ask, assess, evaluate, register, verify-log	All	test_cli

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